

BitForge Project Schedule (based on PRD v0.2)

Planning date: 2026-10-04 | **Submission:** 2027-01-18 | Dates are MM-DD : 10-xx to 12-xx = 2026, 01-xx = 2027. **Roles:** ARCH architect, RTL Verilog designer, VER verification lead (UVM), SW tools, PD physical design, DOC docs, PM project manager. Consecutive IDs with the same plan are grouped; every PRD v0.2 requirement appears exactly once.

1. Milestone calendar

MS	Window	Exit gate (from PRD)
M0	10-05 to 10-18	ISA + regmap v0.1 frozen, repo/CI, flow runs, vPlan skeleton, simulator decided
M1	10-19 to 11-01	Single-lane RTL, assembler, ISS, UVM env (SPI host + pin agents), UART passes
M2	11-02 to 11-22	4 lanes, crossbar, FIFOs, HIF, RAL tests, SPI + I2C, first area/timing report
M3	11-23 to 12-13	DPI scoreboard, random regression, coverage > 85%, assertions + formal, FPGA demo, S-protocols started
M4	12-14 to 01-04	Coverage closure, gate-level pass, P&R closure, feature freeze 01-04
M5	01-05 to 01-18	Docs, final GDS, vPlan sign-off. Internal submit target 01-14 (4-day buffer)

Go/no-go gates: **10-18** (ISA freeze), **11-08** (instruction memory style, lane count), **12-13** (stretch protocols C-items in or out), **01-04** (freeze).

2. Requirement schedule

2.1 Constraints, architecture, ISA

IDs	Start	Finish	MS	Owner	Deliverable / notes
CON-001, 003, 005, 007	10-05	10-18	M0	PM, RTL	Template builds, license, lint rule for Verilog-only RTL
CON-002	10-05	10-18	M0	PD	Parameterized for 8x4; re-check at 01-04
CON-004	10-05	10-18	M0	PD	Area budget baseline (feeds PHY-004)
CON-006	10-05	01-18	M5	PM	Submission
ARC-011	10-05	10-18	M0	ARCH	Module list + per-module spec
ARC-002, 004,	10-19	11-01	M1	RTL	Imem, single clock, reset state

010

ARC-001	10-19	11-22	M1→M2	RTL	1 lane at M1, 4 lanes at M2
ARC-003, 007, 009	11-02	11-22	M2	RTL	Runtime load, FIFOs, event flags
ARC-005	11-09	11-22	M2	RTL	Sync start
ARC-006	11-02	11-22	M2	RTL, PD	Decision 11-08
ARC-008	11-23	12-13	M3	RTL	Timestamp
ISA-018	10-05	10-18	M0	ARCH	<code>isa.yaml</code> frozen
ISA-001 to 010, 017	10-05	11-01	M1	ARCH, RTL	Spec by 10-18, RTL by 11-01
ISA-013	11-02	11-22	M2	RTL	Host IRQ
ISA-011, 012, 016	11-23	12-13	M3	RTL	CALL/RET, oversample, CRC (area permitting)
ISA-014, 015	12-14	01-04	M4	RTL	C: NRZI/stuffing, Manchester (only if 12-13 gate passes)

2.2 I/O, timing, host interface

IDs	Start	Finish	MS	Owner	Deliverable / notes
IO-005, 008	10-19	11-01	M1	RTL	Synchronizer, tristate reset
IO-001 to 004	11-02	11-22	M2	RTL	Crossbar, conflict policy, open drain
IO-006, 007	11-23	12-13	M3	RTL	Glitch filter, inversion
TIM-001	10-19	01-04	M1→M4	PD	Fmax tracked each merge; final at closure
TIM-002, 003, 004, 005	11-02	11-22	M2	RTL, VER	Measured in sim
TIM-006	12-14	01-04	M4	DOC	Per-protocol rate table
HIF-010	10-05	10-18	M0	ARCH	<code>regmap.yaml</code> frozen
HIF-001, 002, 003, 004, 006, 009	11-02	11-22	M2	RTL	SPI slave, regs, burst, IRQ pin
HIF-007	11-23	12-13	M3	RTL	Debug visibility
HIF-005, 008	12-14	01-04	M4	RTL	C: UART boot, breakpoint

2.3 Protocols, differentiators, software, RTL practice

IDs	Start	Finish	MS	Owner	Deliverable / notes
PRO-001	10-19	11-01	M1	SW	UART firmware, all roles
PRO-002, 003	11-02	11-22	M2	SW	SPI, I2C
PRO-004, 005, 006, 011	11-23	12-13	M3	SW	JTAG, SWD, PS/2, multi-lane demo
PRO-007 to 010	12-14	01-04	M4	SW	C: CAN, USB LS, Ethernet, misc; gate 12-13
DIF-007	10-19	11-01	M1	SW	ISS (same as SW-003)
DIF-006	11-02	01-04	M1→M4	SW	Firmware library, grows with PRO
DIF-001, 002, 003, 005	12-14	01-04	M4	SW, RTL	S: capture, auto-baud, LFSR, trigger
DIF-004	12-14	01-04	M4	SW	C: MITM bridge
SW-001	10-12	11-01	M1	SW	Assembler
SW-003	10-12	11-22	M1→M2	SW	ISS 11-01, DPI-C API 11-22
SW-004	11-02	11-22	M2	SW	Python host library
SW-006	10-19	01-04	M1→M4	SW	Per-protocol examples
SW-002	11-23	12-13	M3	SW	Disassembler
SW-005	12-14	01-04	M4	SW	C: timing analyzer
RTL-001, 002, 003, 005	10-05	10-18	M0	RTL	Style guide, bf_defs.vh, lint in CI; enforced after
RTL-004	10-19	11-01	M1	RTL	Single-domain + sync rule check
RTL-006, 007	11-02	11-22	M2	RTL, VER	Build profiles, bound SVA structure
RTL-008	11-02	12-13	M2→M3	RTL	Module headers
RTL-009	11-23	12-13	M3	RTL	FPGA demo (board ordered by 10-31)

2.4 Verification (UVM)

IDs	Start	Finish	MS	Owner	Deliverable / notes
UVM-039	10-05	10-18	M0	VER	Simulator decision + support matrix
UVM-045	10-05	01-18	M0→M5	VER	vPlan skeleton 10-18, sign-off

						01-18
UVM-040	10-12	11-22	M0→M2	VER		CI lint+smoke by 10-18, nightly by 11-22
UVM-001, 003, 004, 005, 006	10-12	11-01	M1	VER		Env skeleton, interfaces, config, logging
UVM-002	10-12	11-22	M1→M2	VER		Full env
UVM-007, 008, 009	10-19	11-01	M1	VER		SPI host, pin, UART agents
UVM-010, 011	11-02	11-22	M2	VER		SPI, I2C partner agents
UVM-014	10-19	11-22	M1→M2	VER		Protocol checkers with each agent
UVM-015	11-02	11-22	M2	VER		RAL + register tests
UVM-021	10-19	11-22	M1→M2	VER		Layered/virtual sequences
UVM-018	10-26	11-22	M1→M2	VER		Protocol scoreboard
UVM-019	11-02	11-22	M2	VER		FIFO scoreboard
UVM-023	10-26	11-22	M1→M2	VER		Directed opcode tests
UVM-024	10-26	12-13	M1→M3	VER		UART (M1), SPI/I2C (M2), error injection (M3)
UVM-029	10-26	12-13	M1→M3	VER		smoke → nightly → full lists
UVM-034	11-02	12-13	M2→M3	VER		SVA modules
UVM-016, 017	11-09	12-13	M2→M3	VER, SW		DPI reference model + lane scoreboard
UVM-022	11-09	12-13	M2→M3	VER		Constrained-random instruction generator
UVM-031, 032	11-09	12-13	M2→M3	VER		Covergroups; closure in M4
UVM-020	11-23	12-13	M3	VER		Timing checker
UVM-025, 026, 027, 028	11-23	12-13	M3	VER		HIF stress, multi-lane, reset, fault tests
UVM-035, 036	11-23	12-13	M3	VER		Cover properties, formal
UVM-041	11-23	01-04	M3→M4	VER		Coverage merge/report
UVM-033	12-14	01-04	M4	VER		Closure: 95% functional, 90% code
UVM-038	12-07	01-04	M3→M4	VER, PD		First netlist run 12-07, nls smoke pass by 01-04

				VER	get_smoke pass by 01-04
UVM-030	12-14	01-04	M4	VER	S/C protocol tests
UVM-044	12-14	01-14	M4→M5	VER	Bring-up tests
UVM-037, 046	12-14	01-04	M4	VER	C: seq. equivalence, mutation
UVM-042, 043	10-19	01-18	M1→M5	VER	Continuous: AI-tagging, bug tracking
UVM-012, 013	11-23	01-04	M3→M4	VER	Agents for S then C protocols

2.5 Physical design and documentation

IDs	Start	Finish	MS	Owner	Deliverable / notes
PHY-004	10-12	01-04	M0→M4	PD	First synthesis 10-18, then each merge
PHY-002	11-23	01-04	M3→M4	PD	First P&R trial 11-23, clean flow 01-04
PHY-006	11-23	12-13	M3	PD	MIN profile fallback proven
PHY-001, 003	12-14	01-04	M4	PD	Area ≤ 65%, timing closed
PHY-005	01-05	01-12	M5	PD	Power estimate
OSS-001	10-05	10-12	M0	PM	Public repo
OSS-004	10-12	11-22	M0→M2	PM, VER	One-command build and regression
OSS-007	11-02	11-22	M2	DOC	Contribution guide
OSS-002	11-02	01-18	M2→M5	DOC	Datasheet draft then final
OSS-005	12-14	01-04	M4	DOC	Dev-board notes
OSS-003, 006	12-14	01-11	M4→M5	DOC, VER	Tutorials and verification guide

3. Critical path

ISA freeze (10-18) → single-lane RTL + assembler + ISS (11-01) → lane scoreboard with DPI model (12-13) → netlist from first stable P&R (12-07) → gate-level pass + coverage closure (01-04) → final GDS + sign-off (01-14).

The risks that can move this path: UVM simulator access (UVM-039, decide by 10-18), instruction-memory area (ARC-006, decide by 11-08), and P&R timing closure (PHY-002/003).

4. Descoping order if behind schedule